

Report of Calibration
PSC ALSu_1ST_ARTICLE_4-CH-FAST_HSFS/N 0001
2026-09-23 15:33:53

Calibration Current standard: BNL PSC ATE S/N 001
Calibration Volt standard: HP 3458A-002 S/N 2823A 23647
Calibration Resistance standard: Fluke 742A-1 S/N 1063008
End Header

lab{2}Chan1
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003477	1.006414	-1.012929	-1.013788	1.005896	-0.002149
-27.000692	27.006624	-26.824911	-26.827932	27.095371	-0.001756

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.002822	-0.016602	-0.017377	0.000000
Initial measured gains:	1.000115	0.992875	0.992958	1.003433
Gain corrections:	1.000115	1.007176	1.007092	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.002332
27.000000	27.094698

Measured offset: -0.001220
Measured gain: 1.003553
Gain correction: 0.996460

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003415	1.003541	-1.003397	-1.003501	1.004098	0.013461
-27.000625	27.001110	-27.000645	-27.000710	27.001575	0.058546

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	-0.000112	0.000020	-0.000086	0.000561
Final measured gains:	1.000014	1.000001	1.000000	0.999996

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000068	0.000077	0.000039	0.000210
Final measured offsets stdev:	0.000223	0.000134	0.000133	0.000318
Final measured gains mean:	1.000005	1.000003	1.000001	0.999988
Final measured gains stdev:	0.000018	0.000008	0.000009	0.000013

Saving channel 1 calibration constants to qspi

lab{2}Chan2
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003131	1.006463	-1.014513	-1.012373	1.003922	0.036747
-27.000492	27.023639	-26.835609	-26.819263	27.095833	0.046158

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.002568	-0.018183	-0.016591	0.000000
Initial measured gains:	1.000762	0.993220	0.992673	1.002873
Gain corrections:	1.000762	1.006826	1.007381	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.000567
27.000000	27.095322

Measured offset: -0.003077
Measured gain: 1.003644
Gain correction: 0.996369

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003017	1.002769	-1.003112	-1.003145	1.003495	-0.004291
-27.000831	27.000453	-27.000412	-27.000406	27.000546	0.054862

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	0.000243	-0.000114	-0.000149	0.000750
Final measured gains:	0.999995	0.999980	0.999979	0.999976

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000045	0.000008	0.000049	-0.000002
Final measured offsets stdev:	0.000254	0.000180	0.000130	0.000617
Final measured gains mean:	1.000004	0.999999	0.999998	0.999997
Final measured gains stdev:	0.000013	0.000013	0.000012	0.000018

Saving channel 2 calibration constants to qspi

lab{2}Chan3
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003092	1.005090	-1.011912	-1.012256	1.008182	0.025176
-27.000691	26.996878	-26.817095	-26.824310	27.082357	0.041725

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.002222	-0.016245	-0.016323	0.000000
Initial measured gains:	0.999776	0.992599	0.992863	1.003170
Gain corrections:	0.999776	1.007457	1.007188	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.005200
27.000000	27.081711

Measured offset: 0.002257
Measured gain: 1.002943
Gain correction: 0.997066

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003103	1.003059	-1.003168	-1.003162	1.003637	0.016584
-27.000723	27.000368	-27.000500	-27.000515	27.001446	0.015909

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	0.000032	-0.000076	-0.000069	0.000559
Final measured gains:	0.999988	0.999989	0.999990	1.000019

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000048	-0.000097	-0.000113	0.000555
Final measured offsets stdev:	0.000109	0.000053	0.000037	0.000052
Final measured gains mean:	0.999994	0.999990	0.999988	1.000015
Final measured gains stdev:	0.000017	0.000011	0.000010	0.000007

Saving channel 3 calibration constants to qspi

lab{2}Chan4
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.002768	1.007532	-1.015173	-1.013833	1.004284	0.020536
-27.000690	26.994422	-26.804020	-26.835772	27.040106	-0.033224

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.005190	-0.020470	-0.017854	0.000000
Initial measured gains:	0.999576	0.991958	0.993231	1.001883
Gain corrections:	0.999576	1.008107	1.006815	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.001258
27.000000	27.039381

Measured offset: -0.000208

Measured gain: 1.001466

Gain correction: 0.998536

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.002839	1.002754	-1.002651	-1.002645	1.003286	0.042847
-27.000392	27.000925	-27.000435	-27.000397	27.001005	0.031298

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	0.000108	0.000197	0.000201	0.000549
Final measured gains:	1.000024	1.000009	1.000008	0.999983

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	-0.000067	-0.000063	-0.000082	0.000283
Final measured offsets stdev:	0.000190	0.000196	0.000190	0.000161
Final measured gains mean:	0.999989	0.999990	0.999987	0.999986
Final measured gains stdev:	0.000022	0.000012	0.000012	0.000007

Saving channel 4 calibration constants to qspi

Test data reviewed by _____ Date_____